

SUNAIN SYED

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Education

University College London (UCL), London

2024 – 2028

Master of Engineering in Computer Science, minoring in Nanotechnology

- Expected First Class Honours (current weighted average: **80.8%**)

Work Experience

Ministry of Housing, Communities and Local Government, Manchester

Jun – Sept 2026

Statistics & Data Analyst Intern | R, Python, Excel

- Automated a data collection via Python APIs, replacing a recurrent manual **1.5 hour workflow to <5 seconds**
- Contributed analytical QA, data visualisation and public communication material for **4 official UK Government statistical releases**
- Conducted an investigation into historical data, producing automated QA R scripts, combining datasets into a larger time series and extending statistical analysis capabilities
- Delivered technical training on API automation and practical LLM adoption to the Housing & Planning Statistics team

Projects

Football Practice - Industry Partner Project (MotionInput Games Ltd) | *C#, Unity* Oct 2025 – Mar 2026

- Delivered a production-ready motion-controlled football training game for the **National Autistic Society** within a five-person team, **culminating in Microsoft Store release**
- Architected a modular shared-core Unity architecture using state machines, dependency inversion, logging and audio services to eliminate cross-module coupling
- Designed and implemented the Freekick minigame end-to-end using a state-driven lifecycle and deterministic physics with controlled randomness for consistent, replayable gameplay
- Developed **accessible UI/UX systems for neurodivergent users** through dynamic menu flows, difficulty scaling, and visual-audio synchronised interactions
- Authored system design documentation and deployment manuals for a live project website; presented to **20+ industry partners** at BettUK 2026 and UCL IXN Showcase

ReelTrust - HackLondon 2026 Winner | *Kotlin, Presage SDK, Node.js, Solana, MongoDB Atlas*

Feb 2026

- Awarded “**Best Use of Presage**” by Major League Hacking for technical innovation and execution at **HackLondon 2026** among **200+ participants**
- Built a full-stack deepfake detection and video verification platform in 24 hours using Kotlin, Node.js, and Solana
- Integrated Presage Android SDK for real-time facial micro-expression analysis, generating authenticity confidence scores for uploaded videos
- Designed a tamper-resistant verification pipeline by anchoring authenticity metadata to Solana and persisting records in MongoDB Atlas

UniverShell | *C, Make*

July 2026

- Developed and **released a cross-platform shell** in C supporting both Unix-like systems and native Windows
- Implemented lexical command parsing, platform-specific process creation and execution through a unified abstraction layer spanning POSIX and Win32 APIs
- Created a system agnostic build system with Make, developing custom macros for conditional builds

Student Grade Management System | *Java, Spring Boot, TypeScript, React, H2 (PostgreSQL)*

Oct – Dec 2025

- Led a 3-person team to develop a full-stack RESTful system for managing student and module records
- Developed a Spring Boot backend with DTO abstractions, global exception handling, and **100% JUnit5 test coverage**
- Created a React + TypeScript frontend with client-side validation, CSV import/export and automated PDF reporting

Extracurriculars

EPQ: Fermat's Last Theorem and Cryptography

Aug 2022 – Apr 2023

'To what extent did the proof of Fermat's Last Theorem influence the field of cryptography?'

- Researched Wiles' proof and elliptic curves, producing a report and presentation on mathematical results, cryptographic applications and quantum computing implications

MEI Ritangle Maths Competition Finalist x2

2021, 2022

- Solved team-based problems using algorithmic approaches: **2022 final**, used ant colony optimisation for fastest 3D terrain routing; **2021 final**, optimised hollow sphere placement - outperformed official solution by 7%

Technical Skills

Programming Languages: Python, Java, C, C#, R, TypeScript, Haskell

Libraries/Frameworks: Spring Boot, React, TensorFlow, NumPy, Pygame, JUnit

Tools: Git, GitHub, PostgreSQL, Docker, Unity, Make, Excel